#ManyDaughters
Pre-analysis plan

Short literature review

The the "daughter effect" is concerned with the question of how the presence of daughters influences individuals' attitudes and behaviors, particularly regarding gender equality and political preferences. There exist many studies concerned with the question, we mention few of them. In one of the first publications, Washington (2008) investigated the influence of child gender, showing that each additional daughter increases a congressperson's likelihood of voting liberally, especially on reproductive rights issues. Sharrow et al. (2018) explored whether having daughters affects men's support for gender-equality policies. Using survey data, they found that having a daughter as a first child significantly increases fathers' support for such policies. However, in their study, having daughters in general or a higher proportion of daughters did not show the same effect. Wu et al. (2023) revealed that male founders with daughters exhibit higher female representation among directors and employees within their ventures. The effect was found to be more pronounced as daughters grow and socialize. Oswald and Powdthavee (2010) presented evidence that having daughters predisposes individuals to vote for left-wing parties, while sons sway preferences towards right-wing parties. Glynn and Sen (2015) utilized judges' family data to show that judges with daughters, especially Republican ones, are more likely to rule in favor of women's issues.

There are only few studies for Germany using the German Socio-Economic Panel (SOEP). Leder & Niszczota (2022) found no significant daughter effect in male or female behavior during a dictator game. Choi et al. (2008) reported that a first-born son increases a father's work hours more than a first-born daughter, particularly among educated fathers. Pollmann-Schult (2015) found that both mothers and fathers work more hours compared to parents of daughters, with this effect being stronger for mothers. Additionally, he documented that having daughters is linked to more traditional labor divisions than sons, with parents spending slightly more time on household tasks when they have children of the same sex.

Threats to causality and approaches to estimating the daughter effect

In estimating the daughter effect, it is important to recognize potential threats to causality, such as differing parental preferences for the gender of their children resulting in different composition of offspring than would occur naturally. The literature has adopted various approaches to study the daughter effect. Some studies look at the effect of the gender of the first-born to avoid the “stopping rule” (Wu et al. 2023; Oswald and Powdthavee 2010) or restrict the sample to parents with only one child (Wu et al. 2023). Other studies control for the total numbers of children (for example, Washington 2008). Another common approach involves examining families with at least one daughter (independently of the order) and including comparison groups that may also have childless individuals (Stöckel et al 2025; Oswald & Powdthavee 2010; Cronqvist & Yu, 2017). Other studies utilizes the count or proportion of daughters as a variable. As pointed out earlier, not all aproaches yield the same results.

Regarding child sex preferences in Germany, studies suggest parents of daughters are more likely to have another child and more likely to divorce compared to parents of sons (Brockmann 2001; Hank and Kohler 2003; Diekmann and Schmidheiny 2004). However, these preferences seem to have decreased over time, losing their importance in contemporary Germany. However, those average trends may mask some heterogeneity, e.g. with more liberal individuals

potentially preferring daughters and conservatives preferring sons, still posing a challenge to causal identification.

The potential other problem of selective abortion based on sex is highly unlikely in Germany due to legal restrictions. The Embryonenschutzgesetz: ESchG §3 (Embryo Protection Act) imposes punishments for sex selection at the fertilization stage. Additionally, the sex of an unborn child can only be detected after 12-14 weeks, surpassing the legal abortion period (up to 12 weeks).

Given this, we believe that only the gender of the first child can be treated as close to random. Therefore, we include only parents with their first child at the defined survey time. Bias may occur if parents of first-born daughters more often decide to have a second child, which might lead to their exclusion from our sample (more traditional families). To account for this and other potential threats, whenever possible, we control for baseline attitudes observed at least a year before the first child was born.

Hypotheses and details of the planned analysis

H1. *Having daughters leads to stronger support for gender equality (as measured by attitudes).*

That is, denoting X as a measure of support for gender equality,
(X for respondents with daughters) > (X for respondents without daughters).

main variables	*gender equality (as measured by attitudes)* items [2012, 2018] plh0298_v[1, 2] A child under 6 suffers if the mother works (reverse) plh0302_v[1, 2] A child under 3 suffers if the mother works (reverse) plh0308_v[1, 2] Best if man and woman work the same amount so they can share the responsibility We use the above items as we understand that only those 3 are available in both waves. operationalization We generate an index gender_equality_attitudes_[2012, 2018] as follows: =14-plh0298_v[1, 2] -plh0302_v[1, 2] + 2*plh0308_v[1, 2] and then standardize.
Inclusion/exclusion criteria	The main waves that we consider are 2012 and 2018 due to the availability of gender attitude questions. We only include parents to whom their first child was born between the two waves 2012 and 2018 and who had no further kids during this period. Additionally, we only include parents who live with the child in the same household at the time the 2018 survey was conducted. We exclude individuals who were younger than 18 or older than 42 years of age at the birth of the first child. items pld0174 Birth Year Of First Child pld0172 Number Of Children pld0176 First Child In Household

Exact analysis	Below, we present the exact analysis as Stata code. Column 1: reg gender_equality_attitudes_2018 **daughter** gender_attitude_items_2012 plh*[2012] CONTROLS1, robust Column 2: Column 1: reg gender_equality_attitudes_2018 **daughter** gender_attitude_items_2012 plh*[2012] CONTROLS1 CONTROLS2, robust Column 3: Column 1: reg gender_equality_attitudes_2018 **daughter** gender_attitude_items_2012 plh*[2012] CONTROLS1 CONTROLS2 CONTROLS3, robust Should this approach result in too few observations (less than 200 per child gender, total of 400), we will extend the sample to individuals to whom their first child was born before 2012 and run the following specification: Column 1: reg gender_equality_attutudes_[2012, 2018] **daughter** CONTROLS1 CONTROLS2 CONTROLS3 Variables and operationalization: In the following t means the wave at which the outcomes are measured (2018 or 2012/2018 in case the first approach does not work). t-1 means the baseline wave, before the child was born (2012 for the first approach). We might supplement the variables (non-attitudes, if missing) with values collected at different waves. <u>child gender</u> pld0175 Sex Of First Child -> daughter dummy <u>gender attitudes items 2012</u> plh0301 Women Should Rather Care About Family Than Career plh0303 Men Involved in Housework plh0304 Children Suffer When Father Focused On Career plh0307 Men Care more About Financial Security plh0309 Working Mothers Equal Emotional CONTROLS1 <u>parent gender</u> pl/pla0009_h Sex [harmonised] -> father dummy <u>education</u> plg0037 School-Leaving Degree-> dummies for secondary general school, intermediate school, technical school, upper secondary school plg0014 College Degree -> college degree dummy <u>age</u> pld0174 Birth Year Of First Child –> age child at wave t =t- pld0174 ple0001 Age Of Person Surveyed <u>other:</u> iyear year of interview-> dummies for times t and t-1 CONTROLS2: <u>demographic:</u> plh0258_h Church, Religion [harmonised] -> dummies catholic, protestant, no confession (missings or other religions receive a value zero) migback Migration background -> dummy urban/rural dummy (if available) pld0131_h -> married dummy, partnership dummy at t and at t-1

	marriage /partnership length (if available) hlc0005_v2 Monthly HH net income (Euro) [2002-2023] (if available) plb0022_h Erwerbsstatus [harmonised] -> dummies full time, part-time, not employed plb0186_h Actual Weekly Working Hours [harmonised] -> wisorized at 99 percentile plc0446 Disability -> dummy disable CONTROLS3 <u>socialization:</u> birthregion_ew Birth place: German Federal Land (East-West Version) -> dummy former East territory lb1190 Mother's employment status, when respondent was under 6 years old (if available) lb0067 No. Of Years Living With Single Mother, lb0069 No. Of Years Living With Single Father -> dummy parent living in a single parent hh as child pla0011 Siblings dummy, pla0017-44 1.-10. Sibling, Sex -> total number older/younger sister; total number older/younger brother (missings or no siblings receive a value zero) Use of weights: Since the sample for the analysis is restricted to individuals with one child of a particular age, we do not consider using sample weights, as the sample is not representative anyhow.
Interpretation of results	If the coefficient in Column 1 on the variable **daughter** is positive and statistically significant at $p<0.05$ (two-sided test), we interpret the results in line with our hypothesis. The coefficients on the variable **daugther** in the other columns provide robustness checks.

H2. *Having daughters leads to more left-leaning political preferences.*
That is, denoting Y as a measure of left-leaning political preferences,
(Y for respondents with daughters) > (Y for respondents without daughters).

main variables	*political preferences* <u>item</u> plh0004 Political attitude [2005,2009,2014,2019] <u>operationalization</u> Generate variable polpref_t from reversed item plh0004 and then standardize.
Inclusion/exclusion criteria	We use individuals who answered plh0004 in at least 2 of the waves—2005, 2009, 2014, 2019—and to whom their first child was born between two waves and who had no further kids during this period. Additionally, we only include parents who live with the child in the same household at the time the last survey was conducted. We exclude individuals who were younger than 18 or older than 42 years of age at the birth of the first child.
Exact analysis	Column 1: reg polpref_t **daughter** polpref_t-1 CONTROL1, robust Column 2: reg polpref_t **daughter** polpref_t-1 CONTROL1 CONTROL2, robust Column 3: reg polpref_t **daughter** polpref_t-1 CONTROL1 CONTROL2 CONTROL3, robust

Interpretation of results	If the coefficient in Column 1 on the variable **daughter** is positive and statistically significant at p<0.05 (two-sided test), we interpret the results in line with our hypothesis. The coefficients on the variable **daugther** in the other columns provide a robustness check.

H3. *Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).*
That is, denoting Z as a measure of gender equality in the household,
(Z for respondents with daughters) > (Z for respondents without daughters).

main variables	*gender equality in the household (as measured by behaviors or outcomes)* items [2000-2020] pli0044_h Hours Weekday Child Care pli0019_h Hours Saturday Child Care pli0022_h Hours Sunday Child Care items [2001, 2003, 2005, 2007, 2009, 2011-2013, 2015, 2017, 2019] pli0043_h Hours Weekday Housework pli0012_h Hours Saturday Housework pli0016_h Hours Sunday Housework operationalization 1. Winsorize all variables at max 12h per day 2. Calculate total hours spend weekly on childcare and housework for both partners separately, where i is the respondent and pi is the partner: Index childcare (sum): 5*pli0044_h+pli0019_h+pli0022_h -> total_childcare_i, total_childcare_pi Index housework (sum): 5*pli0043_h+pli0012_h+pli0016_h -> total_housework_i, total_housework_pi Index total_domesticwork: total_childcare_[i, pi] + total_housework_[i, pi] -> total_domesticwork[i, pi] 3. share_domesticwork_i=total_domesticwork_i/(total_domesticwork_i+ total_domesticwork_pi) share_housework_i=total_housework_i/(total_housework_i+ total_housework_pi)
Inclusion/exclusion criteria	We use individuals who answered pli* in at least 2 of the waves 2001, 2003, 2005, 2007, 2009, 2011-2013, 2015, 2017, 2019 and to whom their first child was born between those two waves and who have no more children during this period. We only include parents who live with the child at the time of the more recent of the two waves. We include only families with both mother and father and living together. We exclude individuals who were younger than 18 or older than 42 years of age at the birth of the first child. items lb0646: Lives Together With Partner [B] lb0658: Lives Together With Partner [C] lb0671: Lives Together With Partner [D]

Exact analysis	Column 1: reg share_domesticwork_t **daughter** share_domesticwork_t-1 father CONTROLS1, cluster(hh_id) Column 2: reg share_domesticwork_t **daughter** share_domesticwork_t-1 father CONTROLS1 CONTROLS2, cluster(hh_id) Column 3: reg share_domesticwork_t **daughter** share_domesticwork_t-1 father CONTROLS1 CONTROLS2 CONTROLS3, cluster(hh_id) Column 4: reg share_housework_t **daughter** share_housework_t-1 father CONTROLS1, cluster(hh_id) Column 5: reg share_housework_t **daughter** share_housework_t-1 father CONTROLS1 CONTROLS2, cluster(hh_id) Column 6: reg share_housework_t **daughter** share_housework_t-1 father CONTROLS1 CONTROLS2 CONTROLS3, cluster(hh_id)
Interpretation of results	If the coefficient in Column 1 on the variable **daughter** is positive and statistically significant at $p<0.05$ (two-sided test), we interpret the results in line with our hypothesis. The coefficients on the variable **daugther** in the other columns provide a robustness check.

H4. *Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).*
That is, denoting X as a measure of support for gender equality,
(X for male respondents with daughters − X for male respondents without daughters)>
(X for female respondents with daughters − X for female respondents without daughters)

main variables	*gender equality (as measured by attitudes)* items [2012, 2018] plh0298_v1/2: [de] Fam: Kind unter 6 J. leidet wenn Mutter arbeitet (reverse) plh0302_v1/2 Kind unter 3 J. leidet wenn Mutter arbeitet (reverse) plh0308_v1/2 Mann/Frau in gleich in Arbeit u. Haushalt/Familie We use the above items as we understand that only those 3 are available in both waves. operationalization We generate an index gender_equality_attitudes_[2012, 2018] as follows: =14-plh0298_v[1, 2] -plh0302_v[1, 2] + 2*plh0308_v[1, 2] and then standardize.
Inclusion/exclusion criteria	The main waves that we consider are 2012 and 2018 due to the availability of gender attitude questions. We only include parents to whom their first child was born between the two waves 2012 and 2018 and who had no further kids during this period. Additionally, we only include parents who live with the child in the same household at the time the 2018 survey was conducted. We exclude individuals who were younger than 18 or older than 42 years of age at the birth of the first child.
Exact analysis	Column 1: reg gender_equality_attitudes_2018 daughter **daughter*father** gender_equality_attitudes_2012 plh*[2012] CONTROLS1, robust Column 2: reg gender_equality_attitudes_2018 daughter **daughter*father** gender_equality_attitudes_2012 plh*[2012] CONTROLS1 CONTROLS2, robust Column 3: reg gender_equality_attitudes_2018 daughter **daughter*father** gender_equality_attitudes_2012 plh*[2012] CONTROLS1 CONTROLS2 CONTROLS3, robust

	Should this approach result in too few observations (less than 200 per child gender), we will extend the sample to individuals to whom their first child was born before 2012 and run the following specification Column 1: reg gender_equality_attitudes_[2012,2018] daughter **daughter*father** CONTROLS1 CONTROLS2 CONTROLS3, robust
Interpretation of results	If the coefficient in Column 1 on the interaction variable **daughter*father** is positive and statistically significant at $p<0.05$ (two-sided test), we interpret the results in line with our hypothesis. The coefficients on the interaction variable **daughter*father** in the other columns provide a robustness check.

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